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Air Handling Unit Construction

Abstract

An air handling unit includes a housing constructed from a plurality of insulation panels embedded with frame elements. The frame elements include at least a first frame element and a second frame element joined together by an inline connector. Each insulation panel includes an outer surface and an opposed inner surface, wherein the insulation panels all cooperate to form a common outer surface of the air handling unit formed of the outer surfaces of the insulation panels and a common inner surface of the air handling unit formed of the inner surfaces of the insulation panels. The common inner surface bounds and defines an interior space of the air handling unit. The frame elements are all inboard of the common outer surface, recessed and embedded within the insulation panels.

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Background/Summary

FIELD

[0001] The present invention relates generally to construction techniques, and more particularly to thermal insulation and air handling unit construction techniques.

BACKGROUND

[0002] Buildings, warehouses, factories, housings, portable structures, and other facilities have different requirements for conditioning their air. Data centers require cooled air. EV battery manufacturing facilities need extremely low humidity levels. Indoor agricultural sites have a variety of environmental needs.

[0003] Specialized industries have specific demands for their air and turn to specially designed air handling units to condition the air. These units may be large and stand alone to house the machinery, servers, or other working components, or they may be appurtenant to a factory or warehouse. Air handling units are typically highly engineered to filter environmental air and condition it as needed. To maintain control, the walls of the units are usually heavily insulated.

[0004] Conventionally, most air handling units for industry use a two-inch thick wall. The walls are usually assembled in one of two ways. Like stick-building construction methods for homes, some units are built out with frames first, and insulation is then placed outside of the frames. In another method of construction, frame elements and insulation panels couple to form a monocoque-like structure.

[0005] The frame elements are usually limited in length. The size of a cargo container is often the constraint factor; frame elements can only be so big so as to still fit onto a container ship for transportation. For example, usually, frame elements cannot be big longer than eighteen feet. This means that, when frame elements are assembled into an air handling unit, the unit can not be longer than eighteen feet.

[0006] This often results in compartmentalized construction of the air handling unit and/or exposed frame elements, both of which reduce the efficacy and efficiency of the unit. For example, many large air handling units are actually built in two or three separate, free-standing modules and then coupled together to form one larger assembled unit. Such units can have boxy construction, complicated structural engineering requirements, leakage, and poor air conditioning performance characteristics.

[0007] Moreover, some customers require single-compartment air handling units. Where air must move through many filtering and conditioning components without interruption by solid walls, a modular construction technique cannot be used. Interior spaces that are segmented by structural elements—such as where two air handling unit housings are brought together to form one larger unit—may not meet a customer's performance requirements. However, there currently is no construction technique that allows for large air handling units with single, continuous, unbroken interiors.

[0008] Still further, some industries require air units which perform so precisely that the conventional two-inch wall is not sufficient. In such situations, a four-inch wall is appropriate. However, no existing construction method or equipment accommodates a four-inch thick panel. There is a need for an improved construction of air handling units.

SUMMARY

[0009] In an embodiment, an air handling unit includes a frame element, and first and second insulation panels which confront the frame element at a juncture. The first insulation panel includes a first distal surface, a first proximal surface, a first intermediate surface extending between the first distal and proximal surfaces, and a first inner surface. The second insulation panel includes a second distal surface, a second proximal surface, a second intermediate surface extending between the second distal and proximal surfaces, and a second inner surface. At the juncture, the second distal surface faces the first intermediate surface at an insulation seam, the second proximal surface faces in spaced relation the first intermediate surface and the second intermediate surface faces in

spaced relation the first proximal surface, thereby defining a closed void defined between the first intermediate surface, the first proximal surface, the second intermediate surface, and the second proximal surface; and the frame element occupies the closed void. A first air-impermeable seal is in the insulation seam, second air-impermeable seals are between the frame element and the first and second intermediate surfaces, respectively, and between the frame element and the first and second inner surfaces, respectively, and third air-impermeable seals are between the frame element and the first and second proximal surfaces, respectively.

[0010] In some embodiments of an air handling unit, the first distal surface is parallel to the first proximal surface, and the second distal surface is parallel to the second proximal surface. The first intermediate surface is normal to the first distal surface and to the first proximal surface, and the second intermediate surface is normal to the second distal surface and to the second proximal surface; and the first inner surface is normal to the first distal surface and to the first proximal surface, and the second inner surface is normal to the second distal surface and to the second proximal surface.

[0011] In some embodiments of an air handling unit, the first and second insulation panels are normal to each other. At the juncture, the first insulation panel overlaps the second insulation panel.

[0012] In some embodiments of an air handling unit, the first air-impermeable seal in the insulation seam is a sealant extending along the insulation seam. The second air-impermeable seal between the frame element and the first intermediate surface includes two gaskets each extending along a length of the frame element. The second air-impermeable seal between the frame element and the second intermediate surface includes another two gaskets each extending along the length of the frame element. The second air-impermeable seal between the frame element and the first inner surface includes another gasket extending along the length of the frame element. The second air-impermeable seal between the frame element and the second inner surface includes yet another gasket extending along the length of the frame element. The third air-impermeable seal between the frame element and the first proximal surface is another sealant extending along a length of the frame element. The third air-impermeable seal between the frame element and the second proximal surface is yet another sealant extending along the length of the frame element.

[0013] In an embodiment, an air handling unit includes a frame element, and first and second insulation panels which confront the frame element at a juncture. The first insulation panel includes a first distal surface, a first proximal surface, a first intermediate surface extending between the first distal and proximal surfaces, and a first inner surface. The second insulation panel includes a second distal surface, a second proximal surface, a second intermediate surface extending between the second distal and proximal surfaces, and a second inner surface. At the juncture, the first distal surface faces the second distal surface at an insulation seam, the first intermediate surface is registered in alignment with the second intermediate surface, the first proximal surface faces the second proximal surface in spaced relation, thereby defining an open void between the first intermediate surface, the first proximal surface, the second intermediate surface, and the second proximal surface. The frame element occupies the open void. A first air-impermeable seal is in the insulation seam, second air-impermeable seals are between the frame element and the first and second intermediate surfaces, respectively, and between the frame element and the first second inner surfaces, respectively, and third air-impermeable seals are between the frame element and the first and second proximal surfaces, respectively.

[0014] In some embodiments of an air handling unit, the first distal surface is parallel to the first proximal surface, and the second distal surface is parallel to the second proximal surface. The first intermediate surface is normal to the first distal surface and to the first proximal surface, and the second intermediate surface is normal to the second distal surface and to the second proximal surface. The first inner surface is normal to the first distal surface and to the first proximal surface, and the second inner surface is normal to the second distal surface and to the second proximal surface.

[0015] In some embodiments of an air handling unit, the first and second insulation panels are registered in alignment with each other, and at the juncture, the first insulation panel abuts the second insulation panel.

[0016] In some embodiments of an air handling unit, the first air-impermeable seal in the insulation seam is a sealant extending along the insulation seam. The second air-impermeable seal between the frame element and the first intermediate surface includes two gaskets each extending along a length of the frame element. The second air-impermeable seal between the frame element and the second intermediate surface includes another two gaskets each extending along the length of the frame element. The second air-impermeable seal between the frame element and the first inner surface includes another gasket extending along a length of the frame element, the second air-impermeable seal between the frame element and the second inner surface includes yet another gasket extending along the length of the frame element, the third air-impermeable seal between the frame element and the first proximal surface is another sealant extending a length of the frame element, and the third air-impermeable seal between the frame element and the second proximal surface is yet another sealant extending along the length of the frame element.

[0017] In an embodiment, an air handling unit includes a housing constructed from a plurality of insulation panels embedded with frame elements. The frame elements include at least a first frame element and a second frame element joined together by an inline connector. Each insulation panel includes an outer surface and an opposed inner surface, wherein the insulation panels all cooperate to form (1) a common outer surface of the air handling unit included of the outer surfaces of the insulation panels and (2) a common inner surface of the air handling unit included of the inner surfaces of the insulation panels. The common inner surface bounds and defines an interior space of the air handling unit. The frame elements are all inboard of the common outer surface, recessed and embedded within the insulation panels.

[0018] In some embodiments of an air handling unit, the inline connector includes a middle section, a first mating portion extending from the middle section in a first direction to a first end, a second mating portion extending from the middle section in a second direction, opposite the first direction, to a second end, the first mating portion including a first major plug and a first minor plug, wherein the first major and minor plugs are configured to fit into sockets in the frame elements, the second mating portion including a second major plug and a second minor plug, wherein the second major and minor plugs are configured to fit into the sockets in the frame elements, and the middle section includes a generally rectangular prismatic body having a dimension larger than the first and second mating portions, and two flanges projecting outwardly away from the body.

[0019] In some embodiments of an air handling unit, the first major plug and the first minor plug are separated by a slender first gap. The second major plug and the second minor plug are separated by a slender second gap.

[0020] In some embodiments of an air handling unit, the flanges are proximate the first and second minor plugs.

[0021] In some embodiments of an air handling unit, the insulation panels include at least a first insulation panel and a second insulation panel confronting one of the frame elements at a juncture. The first insulation panel includes a first distal surface, a first proximal surface, a first intermediate surface extending between the first distal and proximal surfaces, and a first inner surface. The second insulation panel includes a second distal surface, a second proximal surface, a second intermediate surface extending between the second distal and proximal surfaces, and a second inner surface. At the juncture, the second distal surface faces the first intermediate surface at an insulation seam. The second proximal surface faces in spaced relation the first intermediate surface and the second intermediate surface faces in spaced relation the first proximal surface, thereby defining a closed void between the first intermediate surface, the first proximal surface, the second intermediate surface, and the second proximal surface. At least one of the first and second frame

elements occupies the closed void.

[0022] In some embodiments of an air handling unit, a first air-impermeable seal is in the insulation seam. Second air-impermeable seals are between the frame element and the first and second intermediate surfaces, respectively, and between the frame element and the first and second inner surfaces, respectively. Third air-impermeable seals are between the frame element and the first and second proximal surfaces, respectively.

[0023] In some embodiments of an air handling unit, the first air-impermeable seal in the insulation seam is a sealant extending along the insulation seam. The second air-impermeable seal between the frame element and the first intermediate surface includes two gaskets each extending along a length of the frame element. The second air-impermeable seal between the frame element and the second intermediate surface includes another two gaskets each extending along the length of the frame element. The second air-impermeable seal between the frame element and the first inner surface includes another gasket extending along the length of the frame element. The second air-impermeable seal between the frame element and the second inner surface includes yet another gasket extending along the length of the frame element. The third air-impermeable seal between the frame element and the first proximal surface is another sealant extending along a length of the frame element. The third air-impermeable seal between the frame element and the second proximal surface is yet another sealant extending along the length of the frame element.

[0024] In some embodiments of an air handling unit, the first distal surface is parallel to the first proximal surface, and the second distal surface is parallel to the second proximal surface. The first intermediate surface is normal to the first distal surface and to the first proximal surface, and the second intermediate surface is normal to the second distal surface and to the second proximal surface. The first inner surface is normal to the first distal surface and to the first proximal surface, and the second inner surface is normal to the second distal surface and to the second proximal surface.

[0025] In some embodiments of an air handling unit, the insulation panels include at least a first insulation panel and a second insulation panel confronting one of the frame elements at a juncture. The first insulation panel includes a first distal surface, a first proximal surface, a first intermediate surface extending between the first distal and proximal surfaces, and a first inner surface. The second insulation panel includes a second distal surface, a second proximal surface, a second intermediate surface extending between the second distal and proximal surfaces, and a second inner surface. At the juncture: the first distal surface faces the second distal surface at an insulation seam. The first intermediate surface is registered in alignment with the second intermediate surface. The first proximal surface faces the second proximal surface in spaced relation, thereby defining an open void between the first intermediate surface, the first proximal surface, the second intermediate surface, and the second proximal surface. At least one of the first and second frame elements occupies the open void.

[0026] In some embodiments of an air handling unit, a first air-impermeable seal is in the insulation seam. Second air-impermeable seals are between the frame element and the first and second intermediate surfaces, respectively, and between the frame element and the first second inner surfaces, respectively. Third air-impermeable seals are between the frame element and the first and second proximal surfaces, respectively.

[0027] In some embodiments of an air handling unit, the first air-impermeable seal in the insulation seam is a sealant extending along the insulation seam. The second air-impermeable seal between the frame element and the first intermediate surface includes two gaskets each extending along a length of the frame element. The second air-impermeable seal between the frame element and the second intermediate surface includes another two gaskets each extending along the length of the frame element. The second air-impermeable seal between the frame element and the first inner surface includes another gasket extending along a length of the frame element. The second air-impermeable seal between the frame element and the second inner surface includes yet another

gasket extending along the length of the frame element. The third air-impermeable seal between the frame element and the first proximal surface is another sealant extending a length of the frame element, and the third air-impermeable seal between the frame element and the second proximal surface is yet another sealant extending along the length of the frame element.

[0028] In some embodiments of an air handling unit, the first distal surface is parallel to the first proximal surface, and the second distal surface is parallel to the second proximal surface. The first intermediate surface is normal to the first distal surface and to the first proximal surface, and the second intermediate surface is normal to the second distal surface and to the second proximal surface. The first inner surface is normal to the first distal surface and to the first proximal surface, and the second inner surface is normal to the second distal surface and to the second proximal surface.

[0029] The above provides the reader with a very brief summary of some embodiments described below. Simplifications and omissions are made, and the summary is not intended to limit or define in any way the disclosure. Rather, this brief summary merely introduces the reader to some aspects of some embodiments in preparation for the detailed description that follows.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0030] Referring to the drawings:

[0031] FIG. 1 is a front perspective view of an air handling unit;

[0032] FIG. 2 is a section view taken along the line 2-2 of FIG. 1 showing interiors of the air handling unit;

[0033] FIG. 3 is an enlarged front perspective view of the air handling unit with insulation panels removed to show frame elements of the air handling unit;

[0034] FIG. 4 is an enlarged view of FIG. 3;

[0035] FIGS. 5-7 are views of an inline connector for coupling the frame elements;

[0036] FIG. 8 is a section view taken along the line 8-8 of FIG. 2 showing insulation panels meeting and confronting a frame element at a juncture;

[0037] FIG. 9 is an enlarged view of FIG. 8; and

[0038] FIG. 10 is a section view taken along the line 10-10 of FIG. 2 showing insulation panels meeting and confronting a frame element at another juncture.

DETAILED DESCRIPTION

[0039] Reference now is made to the drawings, in which the same reference characters are used throughout the different figures to designate the same elements. Briefly, the embodiments presented herein are preferred exemplary embodiments and are not intended to limit the scope, applicability, or configuration of all possible embodiments, but rather to provide an enabling description for all possible embodiments within the scope and spirit of the specification. Description of these preferred embodiments is generally made with the use of verbs such as “is” and “are” rather than “may,” “could,” “includes,” “comprises,” and the like, because the description is made with reference to the drawings presented. One having ordinary skill in the art will understand that changes may be made in the structure, arrangement, number, and function of elements and features without departing from the scope and spirit of the specification. Further, the description may omit certain information which is readily known to one having ordinary skill in the art to prevent crowding the description with detail which is not necessary for enablement. Indeed, the diction used herein is meant to be readable and informational rather than to delineate and limit the specification; therefore, the scope and spirit of the specification should not be limited by the following description and its language choices.

[0040] FIG. 1 illustrates an air handling unit 10 assembled with a four-inch wall construction

technique that enhances the thermal insulation of an enclosed space within the unit **10**. The air handling unit **10** shown here is exemplary; it illustrates components that can be used to create and assemble smaller, larger, and differently-shaped units. As such, while this specification describes the structural elements, features, and construction techniques of the unit **10** shown in the drawings, the description is not so limited and should be understood to apply to air handling units of other sizes and shapes assembled with the structural elements, features, and construction techniques described herein.

[0041] The unit **10** includes a top **11**, an opposed bottom **12**, a front **13**, an opposed back **14**, and opposed ends **15** and **16**. Throughout this description, reference will be made to these locations both as absolute locations and as directional cues. For example, if a first element is described as being above a second element, the reader should understand that the first element is closer to the top **11** than is the second element.

[0042] As can be seen fairly clearly in FIG. **1**, the unit **10** is constructed with a large plurality of insulation panels **20** (referred to herein as either “insulation panels” or just “panels”), most of which are preferably rectangular. The panels **20** are laid up in abutting relationship with each other to define seams **21** between adjacent panels **20**, such as is shown along the top **11** among several horizontally-arranged panels **20**. The panels **20** are also arranged in abutting relationship with each other to define corners **22** formed between panels **20** which are transverse to each other. Here, corners **22** are formed between panels **20** which are not just transverse but are actually normal to each other; in other embodiments, other angles between panels **20** are present.

[0043] The unit **10** additionally includes other features, such as intake and exhaust vents, power supplies, and access doors. These features are not discussed further but are visible in many of the drawings. This unit **10** is large, at approximately ten feet high from the bottom **12** to the top **11**, ten feet wide from the front **13** to the back **14**, and twenty-five feet long between the opposed ends **15** and **16**. Continuous-compartment construction of this unit **10** using conventional techniques would be impossible.

[0044] FIG. **2** illustrates a section view of the air handling unit **10** taken along the line 2-2 in FIG. **1**. FIG. **2** shows a lower interior **17** and an upper interior **18** of the unit **10**. The two interiors **17** and **18** are separated here by a horizontal surface—a floor of the upper interior **18** and a ceiling of the lower interior **17**.

[0045] However, this separation is purely exemplary to illustrate the possibility of separation if needed. The reader will understand after reviewing this specification that the horizontal floor separating the interiors **17** and **18** could be removed without alteration of the construction and structural characteristics of the unit **10**. The lower interior **17** is continuous from end **15** to end **16**, as is the upper interior **18**. It would not be possible to have two continuous interiors **17** and **18** using conventional construction techniques.

[0046] The unit **10** is constructed with vertical walls and horizontal floors. FIG. **2** shows two walls **23** and three floors **24**. The two walls **23** are at the end **15**, and the three floors **24** are arranged in spaced-apart relation, one above the other, with the top floor **24** at the top **11**.

[0047] Each wall **23** is constructed and assembled as a unitary, continuous piece from insulation panels **20** and frame elements **25** and **27** embedded therein. In other words, one wall **23** extends entirely from the front **13** of the unit **10** to the back **14**. One wall **23** extends entirely from the end **15** to the end **16**. The walls **23** are constructed from multiple frame elements **25** and **27** and insulation panels **20**, but together, those parts cooperate to form a single, continuous wall **23**. These walls **23** can be extremely long, effectively without limit to length. This enables building of the unit **20** with continuous compartments. During assembly, entirely complete walls **23** are raised and coupled to each other and to the floors **24**, rather than assembling the walls and floors piecemeal or as separate components.

[0048] Similarly, each floor **24** is constructed and assembled as a unitary, continuous piece from insulation panels **20** and frame elements **25** and **27**. In other words, one floor **24** extends entirely

from the front **13** of the unit **10** to the back **14** and from the end **15** to the end **16**. The floors **24** are constructed from multiple frame elements **25** and **27** and insulation panels **20**, but together, those parts cooperate to form a single, continuous floor **24**. The floors **24** can be extremely long, effectively without to length and width. This enables building of the unit **20** with continuous compartments.

[0049] The frame elements **25** and **27** are embedded or nested within the insulation panels **20**. Each insulation panel **20** includes at least one notch which cooperates with a similar notch in an adjacent insulation panel **20** to define a void which receives a frame element **25** or **27**. There, the insulation panels **20** are secured to the frame element **25** or **27**. By embedding the frame elements **25** or **27** between the insulation panels **20** and then connecting the frame elements **25** or **27** to each other, a large wall **23** or floor **24** can be constructed. The size of the resultant wall **23** or floor **24** is determined by the length of the frame elements **25**. The frame elements **25** and can be connected end-to-end to construct assemblies of frame elements **25** and **27** with longer effective lengths.

[0050] FIG. **3** shows an upper portion of the unit **10** with many of the insulation panels removed, so that the frame elements **25** and **27** can be seen. There are many frame elements **25** and **27**, extending both vertically and horizontally. Each of the frame elements **25** is identical to the other, and each of the frame elements **27** are identical to each other. The frame elements **25** and **27** are quite similar, however, and so for much of this description, the reference character **25** is used to describe both. Nonetheless, where appropriate and helpful for understanding, the description separately identifies the frame elements **25** and **27**. The frame elements **25** and **27** cooperate to form a frame **28** of the unit **10**, and when insulation panels **20** are added to that frame **28**, they cooperate to define a housing **29** of the kind shown in FIGS. **1-3**. The housing **29** is a boundary between the environment and the conditioned air in the interiors **17** and **18**.

[0051] The frame elements **25** are coupled to each other with different couplings. At the corner between the top **11** and the end **15**, several vertical frame elements **25** and **27** are coupled to the underside of a horizontal frame element **25** extending along that corner. Similarly, at the corner between the top **11** and the front **13**, several vertical frame elements **25** and **27** are coupled to the underside of a horizontal frame element **25** extending that corner. The top of each vertical frame element **27** is secured with a cap **26**, and the screws extend from the cap **26** into the horizontal frame element. The cap **26** rigidly and securely couples the vertical frame element **27** to the horizontal frame element **25** at that corner. The cap **26** can be seen better in FIG. **4**, which is an enlarged view of FIG. **3** detailing part of an outside corner **30** of the unit **10**.

[0052] At the corner **30**, three frame elements **25** are coupled with a corner connector **31**. Each of the frame elements **25** is coupled rigidly and securely to the corner connector **31**. Elements such as the cap **26** and the corner connector **31** enable the walls **23** and floors **24** to be built to accommodate the insulation panels **20**. By spacing apart the intermediate frame elements **27**—the frame elements **27** between the frame elements **25** at the ends or sides of the wall **23** or floor **24**, holds **32** are created to receive the insulation panels **20**, which can be placed precisely into the holds **32** with a snug fit between the frame elements **25**.

[0053] The walls **23** and floors **24** may be built as long as needed by extending the length of some of the frame elements **25**. Referring now to FIG. **4**, an inline connector **33** is shown between two frame elements **25**. The inline connector **33** has a body **40** with a middle section **41** and two mating portions **42** and **43** which extend from the middle section **41** in opposite directions from each other. FIG. **4** shows only the middle section **41**, as the two mating portions **42** and **43** receive the adjacent frame elements **25** and are hidden by them. FIGS. **5-7** show the inline connector **33** in additional detail.

[0054] The inline connector **33** has a first end **44**, visible in FIG. **5**, and an opposed second end **45**, hidden in FIG. **5** but identical to the first end **44**. The second end **45** is visible in FIG. **7**, which shows the inline connector **33** from a perspective roughly opposite that of FIG. **5**. The middle section **41** of the body **40** is disposed intermediate the first and second ends **44** and **45**. The middle

section **41** is generally rectangular prismatic having opposed first and second faces **46** and **47** and a prismatic sidewall **48** extending therebetween. The prismatic sidewall **48** includes four major side faces **50**. Chamfers between adjacent side faces **50** form very slender, minor, oblique faces **51**. [0055] As shown best in FIG. 5, two of the major side faces **50** meet at an inside corner **52** from which two flanges **53** and **54** project outwardly away from the body **40**. The flanges **53** and **54** are identical in every respect except location and orientation. Each extends from the inside corner **52** normal to the major side face **50** to which it is adjacent. The flanges **53** and **54** are thus normal to each other. The flanges **53** and **54** are thin.

[0056] Each flange **53** and **54** has a first or inner face **55** and an opposed second or outer face **56**. Between these faces **55** and **56**, each flange **53** and **54** has a small thickness. Each of the flanges **53** and **54** has first and second ends **57** and **58**. The flanges **53** and **54** are coextensive to the middle section **41** between its first and second faces **46** and **47**. In other words, the flanges **53** and **54** preferably terminate at the first and second ends **57** and **58** no further or closer to the first and second ends **44** and **45** than the first and second faces **46** and **47**, respectively. The first and second ends **57** and **58** of the flanges **53** and **54** are thus flush with the first and second faces **46** and **47** of the middle section **41**.

[0057] The middle section **41** and the flanges **53** and **54** are a single, unitary, monolithic structure. The middle section **41** is preferably solid. Extending from the middle section as unitary monolithic extensions are the first and second mating portions **42** and **43** which are also preferably solid. The entire inline connection **33** is constructed of a material or combination of materials which have material characteristics of durability, ruggedness, rigidity, strength, thermal insulation, and which can withstand extremely high temperatures without altering these material characteristics and without deforming.

[0058] The first and second mating portions **42** and **43** are identical in every respect except location and orientation. As such, this specification will primarily describe the first mating portion **42**, with the understanding that the description applies equally to the second mating portion **43** unless otherwise stated. Moreover, the reference characters used to identify the structural elements and features of the first mating portion **42** are also used to identify those same structural elements and features of the second mating portion **43** but are marked with a prime ("'") symbol on the second mating portion **43** to distinguish them.

[0059] The first mating portion **42** includes a major plug **60** and a minor plug **61**, spaced apart by a slender gap **62**. The major plug **60** extends outwardly from the first face **46**. The major plug **60** includes a top **63**, an adjacent side **64**, and a diagonal bottom **65** extending between the top **63** and bottom **65**. Minor faces **67** and **68** are formed as chamfers between the diagonal bottom **65** and the top **63** and side **64**, respectively. The top **63** includes two faces oriented slightly transverse with respect to each other. Similarly, the side **64** includes oriented slightly transverse with respect to each other. The diagonal bottom **65** is a single flat face. The adjacent faces of the top **63** and side **64** are normal to each other. All of the faces of the top **63**, side **64**, and bottom **65** the major plug **60** are normal to the first face **46**. In some embodiments, these faces taper slightly from the first face **46** to the first end **44**. The major plug **60** terminates at the first end **44** with a flat endwall **66**.

[0060] The minor plug **61** extends outwardly from the first face **46**. The minor plug includes a bottom **70**, an adjacent side **71**, and a diagonal top **72** extending between the bottom **70** and side **71**. The bottom **70** has a single flat face, the side **71** has a single flat face, and the diagonal top **72** has a single flat face. Minor faces **73** and **74** are formed as chamfers between the diagonal top **72** and the bottom **70** and side **71**, respectively. The bottom **70** and side **71** are normal to each other. All of the faces of the bottom **70**, side **71**, and top **72** are normal to the first face **46**. In some embodiments, these faces taper slightly from the first face **46** to the first end **44**. The minor plug **61** terminates at the first end **44** with a flat endwall **75**.

[0061] The major and minor plugs **60** and **61** are spaced apart from each other. The bottom **65** of the major plug **60** and the top **72** of the minor plug **61** are parallel and spaced apart from each other

by the gap **62**. Together, the major and minor plugs **60** and **61** occupy nearly the entirety of the first face **46**, leaving only a small confrontation surface **76** of the first face **46** not consumed by the major and minor plugs **60** and **61**. The body **40** has an outer dimension which is larger than the mating portions **42** and **43**, but only slightly larger, such that the confrontation surface **76** is relatively thin, but at least large enough to encounter and confront a thin metal sidewall and internal rib of a frame element **25**.

[0062] As shown in FIG. 5, the inline connector **33** fits between and receives two adjacent frame elements **25**. The frame elements **25** each have a major socket **80** and a minor socket **81** which are open at the ends **82** of the frame elements **25** and which are separated by a thin rib. These sockets **80** and **81** correspond to the shapes and sizes of the inline connector **33**. To connect the frame elements **25**, the first mating portion **42** is pressed into the end **82** of one frame element **25**, and the second mating portion **43** is pressed into the end **82** of the other frame element **25**. The major plugs **60** are press-fit into the major sockets **80** and the minor plugs **61** are press-fit into the minor sockets **81** until the ends **82** of the frame elements **25** encounter in abutment the first and second faces **46** and **47** of the middle section **41**. The flanges **53** and **54** are proximate the minor plugs **60** and encounter corresponding flanges on the frame elements **25**.

[0063] When so arranged, the mating portions **42** and **43** are entirely within the frame elements **25**, only the middle section **41** of the inline connector **33** is visible, and the frame elements **25** are separated by the middle section **41**, as shown in FIG. 4. This couples the adjacent frame elements **25** together in registered alignment and creates a strong connection between them. The frame elements **25** now form a much longer assembly of frame elements **25**. Additional frame elements **25** can be coupled to this assembly in a similar fashion to construct a frame element **25** assembly that extends along the entire length of the unit **10** from end **15** to end **16**.

[0064] Constructing long assemblies of frame elements **25** allows a large continuous interior space to be built. However, transverse frame elements **25** must still span between these long assemblies to define the holds **32** into which the insulation panels **20** are fit. FIGS. 8 and 9 illustrate how the insulation panels **20** are embedded and nested with the frame elements **25**.

[0065] FIGS. 8 and 9 are section views taken along the line 8-8 in FIG. 2. They show a juncture **85** where two insulation panels **20** confront a frame element **25** at a corner **22** of the unit **10**. FIG. 8 shows the juncture **85** and surrounding structure, while FIG. 9 shows the frame element **25** in greater detail. Certain features are not shown in detail in FIG. 8 but are shown in FIG. 9, and vice versa. One insulation panel **20** is arranged horizontally and one insulation panel **20** is arranged vertically. For clarity of the ensuing description, and with respect to FIGS. 8 and 9 only, the horizontal insulation panel is identified with the reference character **20** but the vertical insulation panel is identified with the reference character **20'**.

[0066] The insulation panel **20** includes an outer skin **90** or sheet of metal, such as galvanized stainless steel. The skin **90** wraps around the insulation panel **20**, bounding and encasing an interior **91**. The interior **91** is filled with insulation **92**, shown in FIGS. 8 and 9 as a large plurality of dots. The insulation **92** is preferably expanding foam that expands to fill the entire interior **91**.

[0067] The insulation panel **20** has opposed ends **93** and **94** and opposed sides **95** and **96** extending between those ends **93** and **94** (see FIGS. 1 and 2). The insulation panel **20** also has an outer surface **97** and an opposed inner surface **98**. Between the outer and inner surfaces **97** and **98**, the panel **20** preferably has a thickness of four inches. The side **95** of the panel **20** has a stepped profile. It includes a distal surface **100** adjacent to the outer surface **97**. The distal surface **100** turns down from the outer surface **97** and is preferably normal thereto.

[0068] The distal surface **100** depends from the outer surface **97** approximately two inches, or half the thickness of the panel **20**. There, an intermediate surface **101** extends inboard from the distal surface **100**. The intermediate surface **101** extends inboard approximately four inches. The intermediate surface **101** is preferably normal to the distal surface **100** and is preferably parallel to the outer surface **97**.

[0069] The intermediate surface **101** terminates inwardly at a proximal surface **102**. The proximal surface **102** turns down, depending from the intermediate surface **101** approximately two inches, or half the thickness of the panel **20**. There, the proximal surface **102** meets and forms an inner corner **103** with the inner surface **98** of the panel **20**. The proximal surface **102** is parallel to the distal surface **100** and is normal to the outer, inner, and intermediate surfaces **97**, **98**, and **101**.

[0070] Similarly, the vertical insulation panel **20'** includes an outer skin **110** or sheet of metal, such as galvanized stainless steel. The skin **110** wraps around the insulation panel **20'**, bounding and encasing an interior **111**. The interior **111** is filled with insulation **112**, shown in FIGS. **8** and **9** as a large plurality of dots. The insulation **112** is preferably expanding foam that expands to fill the entire interior **111**.

[0071] The insulation panel **20'** has opposed ends **113** and **114** and opposed sides **115** and **116** extending between those ends **113** and **114** (see FIG. **2**). The insulation panel **20'** also has an outer surface **117** and an opposed inner surface **118**. Between the outer and inner surfaces **117** and **118**, the panel **20'** preferably has a thickness of four inches. The end **114** of the panel **20'** has a stepped profile. It includes a distal surface **120** adjacent to the outer surface **117**. The distal surface **120** turns down from the outer surface **117** and is preferably normal thereto.

[0072] The distal surface **120** depends from the outer surface **117** approximately two inches, or half the thickness of the panel **20'**. There, an intermediate surface **121** extends inboard from the distal surface **120**. The intermediate surface **121** extends inboard approximately two inches. The intermediate surface **121** is preferably normal to the distal surface **120** and is preferably parallel to the outer surface **117**.

[0073] The intermediate surface **121** terminates inwardly at a proximal surface **122**. The proximal surface **122** turns down, depending from the intermediate surface **121** approximately two inches, or half the thickness of the panel **20'**. There, the proximal surface **122** meets and forms an inner corner **123** with the inner surface **118** of the panel **20'**. The proximal surface **122** is parallel to the distal surface **120** and is normal to the outer, inner, and intermediate surfaces **117**, **118**, and **121**.

[0074] As shown in FIGS. **8** and **9**, the horizontal insulation panel **20** is normal to and overlaps the vertical insulation panel **20'** at the juncture **85**. In other embodiments, this arrangement is reversed. In overlapping arrangement, the insulation panels **20** and **20'** cooperate to define a void **130** between the insulation panels **20** and **20'** at their side **95** and end **114**, respectively. The void **130** has an elongate, rectangular prismatic shape that extends along the full length of the end **114** and the side **95**. The void **130** is closed because it is bounded on all sides by the intermediate surface **101**, the intermediate surface **121**, the proximal surface **102**, and the proximal surface **122**. The void **130** is sized and shaped to closely receive the frame element **25**. The void **130** is inboard of both outer surfaces **97** and **117**, recessed within the insulation panels **20** and **20'**. It is therefore insulated from the environment. When a frame element **25** is disposed in the void **130**, that frame element **25** is received in the void **130** and thus encased or embedded in within the insulation panels, recessed from a common outer surface **131** of the unit **10**.

[0075] When the insulation panels **20** and **20'** overlap in the arrangement as shown in FIGS. **8** and **9**, their outer surfaces **97** and **117** cooperate to form a common outer surface **131** of the air handling unit **10** which is continuous across the entire unit **10** except at insulation seams **133** formed between the insulation panels **20**. These seams **133** are sealed, however, as described below.

[0076] Similarly, the inner surfaces **98** and **118** cooperate to form a common inner surface **132** of the air handling unit **10** which is continuous across the entire unit **10** except where frame elements **27** interrupt it, though those frame elements **27** are well sealed to the adjacent insulation panels **20**. The common inner surface **132** bounds and defines the interiors **17** and **18** of the air handling unit **10**.

[0077] When the insulation panels **20** and **20'** overlap as in the arrangement of FIGS. **8** and **9**, the distal surface **100** of the panel **20** and the outer surface **117** of the panel **20'** are flush, aligned, and contiguous to each other, spaced apart only slightly by the seam **133**. The intermediate surface **101**

and the distal surface **120** face each other and are just slightly apart in spaced relation with each other. The intermediate surface **101** and the proximal surface **122** face each other but are set apart in spaced relation by the void **130**, or approximately two inches. The proximal surface **02** and the intermediate surface **121** face each other and are also set apart in spaced relation by the void **130**, or approximately two inches. The inner surfaces **98** and **118** converge proximate each other at the inner corners **103** and **123**.

[0078] The frame element **25** occupies the void **130**. The frame element **25** is an elongate rectangular prismatic member. In the drawings of this embodiment, the frame element **25** has a generally square cross-section as illustrated in FIG. **8**. The frame element **25** has a sidewall **140** formed of a combination of metal and plastic thermal breaks. The sidewall **140** includes a top wall **141**, a right wall **142**, a bottom wall **143**, and a left wall **144**. These names are selected only to easily identify and refer to the walls shown in FIGS. **8** and **9** and not to limit their description or scope in any way. Each of the walls has an inner face, directed into the frame element **25**, and an opposed outer face, directed out of the frame element **25**. These inner and outer faces are not labeled with reference characters because it should be clear from the drawings and the description where these faces are located.

[0079] The top wall **141** is preferably constructed from metal and extends entirely from the left wall **144** to the right wall **142**. Similarly, the right wall is preferably constructed from metal and extends entirely from the top wall **141** to the bottom wall **143**.

[0080] The bottom wall **142**, however, includes a thermal break **145**. The thermal break **145** extends in the bottom wall **143** from the right wall **142** to approximately halfway between the right and left walls **142** and **144**. The thermal break **145** extends entirely along the length of the frame element **25** and as such, is a thermal discontinuity limiting the transfer of thermal energy through the frame element **25** and from the right wall **142** to the metal portion of the bottom wall **143** specifically. Because the bottom wall **143** extends toward the interior **18** of the air handling unit **10**, this helps prevent transfer of the thermal energy into the interior **18** of the unit **10**.

[0081] The left wall **144** also includes a thermal break **146**. The thermal break **146** extends in the left wall **144** from the top wall **141** to approximately halfway between the top and bottom walls **141** and **143**. The thermal break **146** extends entirely along the length of the frame element **25** and as such, is a thermal discontinuity limiting the transfer of thermal energy through the frame element **25** and from the top wall **141** to the metal portion of the left wall **144** specifically. Because the left wall **144** extends toward the interior **18** of the unit **10**, this helps prevent transfer of the thermal energy into the interior **18** of the unit **10**.

[0082] The thermal breaks **145** and **146** are constructed of a material or combination of materials which have material characteristics of durability, ruggedness, resiliency, thermal insulation, and which can withstand extremely high temperatures without altering these material characteristics and without deforming.

[0083] At the juncture of the bottom and left walls **143** and **144**, two flanges **150** and **151** project outwardly away from the sidewall **140** of the frame element **25**. The flanges **150** and **151** are thin and extend from the sidewall **140** to free ends. The flanges **150** and **151** extend along the entire length of the frame element **25**. The flange **150** has an inner face **152** and an opposed outer face **153**. Likewise, the flange **151** has an inner face **154** and an opposed outer face **154**.

[0084] When the frame element **25** occupies the void **130**, the top wall **141** of the frame element **25** faces the intermediate surface **101** of the insulation panel **20** and is slightly apart therefrom. The right wall **142** faces and is spaced slightly apart from the intermediate surface **121** of the insulation panel **20'**. The bottom wall **143** faces and is spaced slightly apart from the proximal surface **122** of the insulation panel **20'**. The left wall **144** faces and is spaced slightly apart from the proximal surface **102** of the insulation panel **20**. The inner face **152** of the flange **150** faces and is spaced slightly apart from the inner surface **118** of the panel **20'**. The inner face **154** of the flange **151** faces and is spaced slightly apart from the inner surface **98** of the panel **20**. Air impermeable seals are

formed between all of these components.

[0085] Between the top wall **141** of the frame element **25** and the intermediate surface **101** of the insulation panel **20**, there are two gaskets **160**. The gaskets **160** extend entirely along the length of the frame element **25**. The gaskets **160** are constructed of a material or combination of materials which have material characteristics of durability, ruggedness, resiliency, thermal insulation, and which can withstand extremely high temperatures without altering these material characteristics and without deforming. One gasket **160** is proximate the right wall **142**, and the other gasket **160** is proximate the left wall **144**. The gaskets **160** each define an air-impermeable seal. Between the two gaskets **160**, self-tapping screws **161** couple the panel **20** to the frame element **25**. Only one such screw **161** is visible in FIG. 9, but there are many self-tapping screws **161** along the length of the frame element **25**. The screws **161** pass through the skin **90** at the inner surface **98** of the panel **20**, with the head of the screw **161** disposed in the interior **91** and the shank of the screw **161** through the top wall **141** and into the frame element **25**.

[0086] Between the right wall **142** and the intermediate surface **121** of the insulation panel **20'**, there are two gaskets **160**. These gaskets **160** also extend entirely along the length of the frame element **25**. These gaskets **160** are constructed of a material or combination of materials which have material characteristics of durability, ruggedness, resiliency, thermal insulation, and which can withstand extremely high temperatures without altering these material characteristics and without deforming. One gasket **160** is proximate the top wall **141**, and the other gasket **160** is proximate the bottom wall **143**. The gaskets **160** each define an air-impermeable seal. Between the two gaskets **160**, self-tapping screws **161** couple the panel **20'** to the frame element **25**. Only one such screw **161** is visible in FIG. 9, but there are many self-tapping screws **161** along the length of the frame element **25**. The screws **161** pass through the skin **110** at the inner surface **118** of the panel **20'**, with the head of the screw **161** disposed in the interior **111** and the shank of the screw **161** through the top wall **141** and into the frame element **25**.

[0087] Between the bottom wall **143** and the proximal surface **122** of the insulation panel **20'**, there is a sealant **162**. The sealant **162** is a caulking material applied during assembly of the unit **10**. The sealant **162** extends broadly across the outer face of the bottom wall **143** substantially entirely from the right wall **142** to the left wall **144**, covering both the metal portion of the bottom wall **143** and the thermal break **145**. The sealant **162** further extends along the entire length of the frame element **25**. The sealant **162** is an air-impermeable seal.

[0088] A thermal break **163** is also fit into a slot **168** in the proximal surface **122**. The thermal break **163** has an H-shape; it includes a base **164** which is disposed in and fills the space between the bottom wall **143** and the proximal surface **122**, a neck **165** which extends through the slot **168**, and a cap **166** which projects laterally outward from the neck **165**, terminating in upwardly-turned lips **167**. The cap **166** of the thermal break **163** is in the interior **111** of the panel **20'**. The thermal break **163** is constructed of a material or combination of materials which have material characteristics of durability, ruggedness, resiliency, thermal insulation, and which can withstand extremely high temperatures without altering these material characteristics and without deforming.

[0089] Between the left wall **144** and the proximal surface **102** of the insulation panel **20**, there is a sealant **162**. The sealant **162** is a caulking material applied during assembly of the unit **10**. The sealant **162** extends broadly across the outer face of the left wall **144** substantially entirely from the top wall **141** to the bottom wall **143**, covering both the metal portion of the left wall **144** and the thermal break **146**. The sealant **162** further extends along the entire length of the frame element **25**. The sealant **162** is an air-impermeable seal.

[0090] Another thermal break **163** is fit into a slot **169** in the proximal surface **122**. The thermal break **163** is identical to the other; it includes a base **164** which is disposed in and fills the space between the left wall **144** and the proximal surface **102**, a neck **165** which extends through the slot **169**, and a cap **166** which projects laterally outward from the neck **165**, terminating in upwardly-turned lips **167**. The cap **166** of this thermal break **163** is in the interior **91** of the panel **20**. The

thermal break **163** is constructed of a material or combination of materials which have material characteristics of durability, ruggedness, resiliency, thermal insulation, and which can withstand extremely high temperatures without altering these material characteristics and without deforming. [0091] Between the inner face **152** of the flange **150** and the inner surface **118** of the panel **20'** is a gasket **160**, proximate to the bottom wall **143**. This gasket **160** defines an air-impermeable seal. This gasket **160** extends entirely along the length of the flange **150** and is constructed of a material or combination of materials which have material characteristics of durability, ruggedness, resiliency, thermal insulation, and which can withstand extremely high temperatures without altering these material characteristics and without deforming. Proximate to the free end of the flange **150**, a self-tapping screw **161** couples the panel **20'** to the frame element **25**. Only one such screw **161** is visible in FIG. **9**, but there are many self-tapping screws **161** along the length of the flange **150**. The screws **161** pass through the skin **110** at the inner surface **118** of the panel **20'**. The head of the screw **161** is disposed in the interior **18** of the air handling unit **10** against the outer face **153** of the flange **150**, and the shank of the screw **161** passes through the inner surface **118** of the panel **20'** and into the interior **111** of the panel **20'**.

[0092] Between the inner face **154** of the flange **151** and the inner surface **98** of the panel **20** is a gasket **160**, proximate to the left wall **144**. This gasket **160** defines an air-impermeable seal. This gasket **160** extends entirely along the length of the flange **151** and is constructed of a material or combination of materials which have material characteristics of durability, ruggedness, resiliency, thermal insulation, and which can withstand extremely high temperatures without altering these material characteristics and without deforming. Proximate to the free end of the flange **151**, a self-tapping screw **161** couples the panel **20** to the frame element **25**. Only one such screw **161** is visible in FIG. **9**, but there are many self-tapping screws **161** along the length of the flange **151**. The screws **161** pass through the skin **90** at the inner surface **98** of the panel **20**. The head of the screw **161** is disposed in the interior **18** of the air handling unit **10** against the outer face **155** of the flange **151**, and the shank of the screw **161** passes through the inner surface **98** of the panel **20** and into the interior **91** of the panel **20**.

[0093] Finally, between the intermediate surface **101** of the panel **20** and the distal surface **120** of the panel **20'**, in the seam **133**, is a sealant **162**. This sealant **162** defines an air-impermeable seal. The sealant **162** extends broadly across the distal surface **120** entirely from intermediate surface **121** to the outer surface **117**. Therefore, the sealant **162** extends across the entire insulation seam **133**.

[0094] In this way, the insulation panels **20** and **20'** are secured to the frame element **25** and form air-impermeable seals with each other and with the frame element **25**. By doing so, they mitigate the transfer of thermal energy from outside of the air handling unit **10** to the interiors **17** and **18** of the unit **10**.

[0095] FIG. **10** is a section view taken along the line **8-8** in FIG. **2**. While FIGS. **8** and **9** illustrate a juncture **85** where two insulation panels **20** confronted a frame element **27** at a corner **22** of the unit **10**, FIG. **10** shows a juncture **85** where two insulation panels **20** abut each other and confront a frame element **27** at a planar section of the unit **10**, such as where two side-by-side coplanar insulation panels **20** meet. These two insulation panels **20** are registered in alignment with each other. Both insulation panels **20** are arranged horizontally in FIG. **10**, though this description is equally applicable where both insulation panels **20** are arranged vertically. Laying multiple panels **20** out in this abutting, registered fashion allows an air handling unit housing to be constructed with long, continuous, unbroken walls **23** and floors **24**.

[0096] Because the two insulation panels **20** are identical to each other in all respects but for location, for clarity of the ensuing description, and with respect to FIG. **10** only, the insulation panel on the left of the drawing is identified with the reference character **20** and the insulation panel on the right is identified with the reference character **20'**. Moreover, the structural elements and features of the insulation panel **20'** on the right carry the prime ("'") designation to distinguish

them from those of the panel **20** on the left. The description below then often refers only to only one of the panels **20** and **20'** and one set of the reference characters—with or without the prime symbol—to avoid repeating reference characters both with and without the prime symbol. The reader will understand that the description applies equally to both panels **20** and **20'**.

[0097] The insulation panel **20** in FIG. **10** is identical to the insulation panel **20'** in FIGS. **8** and **9**. Therefore, it is unnecessary to repeat a full description of the structural elements and features of the panel **20**, and only a limited discussion is presented below, as the reader will understand that the panel **20** has all of the identical structural elements and features of the panel **20'** from FIGS. **8** and **9** and that the prior description of the panel **20'** from FIGS. **8** and **9** is applicable here. The insulation panel **20** has mirror symmetry, such that its ends **113** and **114** are identical, and the structural elements and features that are located at one end **113** are the same as those at the other end **114**.

[0098] Therefore, as with the insulation panel **20'** of FIGS. **8** and **9**, the insulation panel **20** of FIG. **10** has an outer skin **110** fully bounding and encasing an interior **111** filled with insulation **112**. The insulation panel **20** has opposed ends **113** and **114** and opposed sides **115** and **116**, an outer surface **117** and opposed inner surface **118**, and a thickness of preferably four inches. Each of the ends **113** and **114** has a stepped profile with a distal surface **120** adjacent to the outer surface **117** which extends from the outer surface **117** approximately two inches, or half the thickness of the panel **20**. There, an intermediate surface **121** extends inboard from the distal surface **120** approximately two inches, is preferably normal to the distal surface **120**, and is preferably parallel to the outer surface **117**. The intermediate surface **121** terminates inwardly at a proximal surface **122**. The proximal surface **122** turns, depending from the intermediate surface **121** approximately two inches, or half the thickness of the panel **20** to meet and form an inner corner **123** with the inner surface **118** of the panel **20**. The proximal surface **122** is parallel to the distal surface **120** and is normal to the outer, inner, and intermediate surfaces **117**, **118**, and **121**.

[0099] As shown in FIG. **10**, the two insulation panels **20** and **20'** confront each other in horizontal registration. The insulation panels **20** and **20'** cooperate to define an open void **134** between the insulation panels **20** and **20'** at their ends **114** and **114'**, respectively. The void **134** has an elongate, rectangular prismatic shape that extends along the full length of the ends **114** and **114'**. The void **134** is sized and shaped to closely receive the frame element **27**. The void **134** is inboard of both outer surfaces **97** and **117**, recessed within the insulation panels **20** and **20'**, and is open on its bottom along the inner surfaces **118** and **118'**.

[0100] The void **134** is open because it is bounded on all sides except at the opening along the inner surfaces **118** and **118'**. When a frame element **27** is received within the void **134**, the frame element **27** is recessed away from the common outer surface **131** of the unit **10**, and the frame element **27** is sealed to the surrounding faces of the insulation panels **20** and **20'**. The void **134**, and the frame element **27** embedded therein, are therefore insulated from the environment outside the unit **10**.

[0101] Similarly to the juncture **85** of FIGS. **8** and **9**, the outer surfaces **97** and **97'** in FIG. **10** cooperate to form the common outer surface **131** of the air handling unit **10** which is continuous across the entire unit **10** except at the insulation seams **133** formed between the insulation panels **20**. These seams **133** are sealed, however, as described below.

[0102] When the insulation panels **20** and **20'** are registered in horizontal alignment as in the arrangement of FIG. **10**, the distal surfaces **100** and **100'** face each other registration, coextensive to each other, spaced apart only slightly by the seam **133**. The intermediate surfaces **101** and **101'** are registered in coplanar alignment with each other and are, at their outer ends, just slightly apart in spaced relation from each other. The proximal surfaces **122** and **122'** face each other but are set apart in spaced relation by a rectangular prismatic void **134** having a dimension of approximately two inches. The inner surfaces **118** and **118'** are registered in coplanar alignment with each other and are, at their outer ends, also set apart in spaced relation by the void **134**, or approximately two inches.

[0103] The frame element **27** occupies the void **134**. As noted far above, the frame element **27** is similar to the frame element **25** but is not identical, since it joins to coplanar panels **20** and **20'** at a horizontal juncture **86** rather than to transverse panels at a corner juncture **85**. The frame element **27** is an elongate rectangular prismatic member. In the drawings of this embodiment, the frame element **27** has a generally square cross-section as illustrated in FIG. **10**. The frame element **27** has a sidewall **170** formed of a combination of metal and plastic thermal breaks. The sidewall **170** includes a top wall **171**, a right wall **172**, a bottom wall **173**, and a left wall **174**. As above, these names are selected only to easily identify and refer to the walls shown in FIG. **10** and not to limit their description or scope in any way. Each of the walls has an inner face, directed into the frame element **27**, and an opposed outer face, directed out of the frame element **27**. These inner and outer faces are not labeled with reference characters because it should be clear from the drawings and the description where these faces are located.

[0104] The top wall **171** is preferably constructed from metal and extends entirely from the left wall **174** to the right wall **172**. Similarly, the bottom wall **172** is preferably constructed from metal and extends entirely from the left wall **174** to the right wall **172**.

[0105] The right wall **172**, however, includes a thermal break **175**. The thermal break **175** extends in the right wall **172** from the top wall **171** to approximately halfway between the top and bottom walls **171** and **173**. The thermal break **175** extends entirely along the length of the frame element **27** and as such, is a thermal discontinuity limiting the transfer of thermal energy through the frame element **27** and from the top wall **171** to the metal portion of the right wall **172** specifically. Because the right wall **172** extends toward the interior **18** of the air handling unit **10**, this helps prevent transfer of the thermal energy into the interior **18** of the unit **10**.

[0106] The left wall **174** also includes a thermal break **176**. The thermal break **176** extends in the left wall **174** from the top wall **171** to approximately halfway between the top and bottom walls **171** and **173**. The thermal break **176** extends entirely along the length of the frame element **27** and as such, is a thermal discontinuity limiting the transfer of thermal energy through the frame element **27** and from the top wall **171** to the metal portion of the left wall **174** specifically. Because the left wall **174** extends toward the interior **18** of the unit **10**, this helps prevent transfer of the thermal energy into the interiors **17** and **18** of the unit **10**.

[0107] The thermal breaks **175** and **176** are constructed of a material or combination of materials which have material characteristics of durability, ruggedness, resiliency, thermal insulation, and which can withstand extremely high temperatures without altering these material characteristics and without deforming.

[0108] A flange **180** projects outwardly away from the sidewall **170** of the frame element **27** at the corner formed by the right and bottom walls **172** and **173**. Similarly, a flange **181** projects outwardly away from the sidewall **170** of the frame element **27** at the corner formed by the left and bottom walls **174** and **173**. The flanges **180** and **181** are coplanar to the bottom wall **173** and are extensions of that wall **173** terminating at free ends beyond the sidewall **170**.

[0109] The flanges **180** and **181** are thin, and they extend along the entire length of the frame element **27**. The flange **180** has an inner face **182** and an opposed outer face **183**. Likewise, the flange **181** has an inner face **184** and an opposed outer face **184**.

[0110] When the frame element **27** occupies the void **134**, the top wall **171** of the frame element **27** faces the intermediate surfaces **121** and **121'** of the panels **20** and **20'** and is slightly apart from each. The right wall **172** faces and is spaced slightly apart from the proximal surface **122'** of the panel **20'**. The bottom wall **173** is exposed to the open interior **18** of the air handling unit **10**. The left wall **174** faces and is spaced slightly apart from the proximal surface **122** of the insulation panel **20**. The inner face **182** of the flange **180** faces and is spaced slightly apart from the inner surface **118'** of the panel **20'**. The inner face **184** of the flange **181** faces and is spaced slightly apart from the inner surface **98** of the panel **20**. Air impermeable seals are formed between all of these components.

[0111] Between the top wall **171** of the frame element **27** and the intermediate surface **121** of the insulation panel **20**, and also between the top wall **171** of the frame element **27** and the intermediate surface **121'** of the panel **20'**, there are two gaskets **160**. The gaskets **160** extend entirely along the length of the frame element **27**. The gaskets **160** are constructed of a material or combination of materials which have material characteristics of durability, ruggedness, resiliency, thermal insulation, and which can withstand extremely high temperatures without altering these material characteristics and without deforming.

[0112] One gasket **160** is proximate the right wall **172**, another is proximate the left wall **174**, and there two more located therebetween. The gaskets **160** each define an air-impermeable seal. Between the right-most gasket **160** and the gasket **160** to its left, and between the left-most gasket **160** and the gasket **160** to its right, self-tapping screws **161** couple the panel **20** to the frame element **27**. Only two such screws **161** are visible in FIG. **10**, but there are many self-tapping screws **161** along the length of the frame element **27**. The screws **161** pass through the skins **110** and **110'** at the inner surfaces **118** and **118'** of the panels **20** and **20'**, with the heads of the screws **161** disposed in the interiors **111** and **111'** and the shanks of the screws **161** through the top wall **171** and into the frame element **27**.

[0113] Between the right wall **172** and the proximal surface **122'**, there is a sealant **162**. The sealant **162** is a caulking material applied during assembly of the unit **10**. The sealant **162** extends broadly across the outer face of the right wall **172** substantially entirely from the top wall **171** to the bottom wall **173**, covering both the metal portion of the right wall **173** and the thermal break **175**. The sealant **162** further extends along the entire length of the frame element **27**. The sealant **162** is an air-impermeable seal.

[0114] A thermal break **163** is also fit into a slot **168'** in the proximal surface **122'** of the panel **20'**. The thermal break **163** has an H-shape; it includes a base **164** which is disposed in and fills the space between the right wall **172** and the proximal surface **122'**, a neck **165** which extends through the slot **168'**, and a cap **166** which projects laterally outward from the neck **165**, terminating in upwardly-turned lips **167**. The cap **166** of the thermal break **163** is in the interior **111'** of the panel **20'**. The thermal break **163** is constructed of a material or combination of materials which have material characteristics of durability, ruggedness, resiliency, thermal insulation, and which can withstand extremely high temperatures without altering these material characteristics and without deforming.

[0115] Between the left wall **174** and the proximal surface **122** of the insulation panel **20**, there is a sealant **162**. The sealant **162** is a caulking material applied during assembly of the unit **10**. The sealant **162** extends broadly across the outer face of the left wall **174** substantially entirely from the top wall **171** to the bottom wall **173**, covering both the metal portion of the left wall **174** and the thermal break **176**. The sealant **162** further extends along the entire length of the frame element **27**. The sealant **162** is an air-impermeable seal.

[0116] A thermal break **163** is also fit into a slot **168** in the proximal surface **122** of the panel **20**. The thermal break **163** has an H-shape; it includes a base **164** which is disposed in and fills the space between the left wall **174** and the proximal surface **122**, a neck **165** which extends through the slot **168**, and a cap **166** which projects laterally outward from the neck **165**, terminating in upwardly-turned lips **167**. The cap **166** of the thermal break **163** is in the interior **111** of the panel **20**. The thermal break **163** is constructed of a material or combination of materials which have material characteristics of durability, ruggedness, resiliency, thermal insulation, and which can withstand extremely high temperatures without altering these material characteristics and without deforming.

[0117] Between the inner face **182** of the flange **180** and the inner surface **118'** of the panel **20'** is a gasket **160**, proximate to the bottom wall **173**. This gasket **160** defines an air-impermeable seal. This gasket **160** extends entirely along the length of the flange **180** and is constructed of a material or combination of materials which have material characteristics of durability, ruggedness,

resiliency, thermal insulation, and which can withstand extremely high temperatures without altering these material characteristics and without deforming. Proximate to the free end of the flange **180**, a self-tapping screw **161** couples the panel **20'** to the frame element **27**. Only one such screw **161** is visible in FIG. **10**, but there are many self-tapping screws **161** along the length of the flange **180**. The screws **161** pass through the skin **110'** at the inner surface **118'** of the panel **20'**. The head of the screw **161** is disposed in the interior **18** of the air handling unit **10** against the outer face **183** of the flange **180**, and the shank of the screw **161** passes through the inner surface **118'** of the panel **20'** and into the interior **111** of the panel **20'**.

[0118] Between the inner face **184** of the flange **181** and the inner surface **98** of the panel **20** is a gasket **160**, proximate to the left wall **174**. This gasket **160** defines an air-impermeable seal. This gasket **160** extends entirely along the length of the flange **181** and is constructed of a material or combination of materials which have material characteristics of durability, ruggedness, resiliency, thermal insulation, and which can withstand extremely high temperatures without altering these material characteristics and without deforming. Proximate to the free end of the flange **181**, a self-tapping screw **161** couples the panel **20** to the frame element **27**. Only one such screw **161** is visible in FIG. **10**, but there are many self-tapping screws **161** along the length of the flange **181**. The screws **161** pass through the skin **90** at the inner surface **98** of the panel **20**. The head of the screw **161** is disposed in the interior **18** of the air handling unit **10** against the outer face **155** of the flange **181**, and the shank of the screw **161** passes through the inner surface **98** of the panel **20** and into the interior **91** of the panel **20**.

[0119] Finally, between the distal surface **120** of the panel **20** and the distal surface **120'** of the panel **20'**, in the seam **133**, is a sealant **162**. This sealant **162** defines an air-impermeable seal. The sealant **162** extends broadly across the distal surfaces **120** and **120'** entirely from the outer surfaces **117** and **117'** to the intermediate surfaces **121** and **122'**, respectively.

[0120] In this way, the insulation panels **20** and **20'** are secured to the frame element **27** and form air-impermeable seals with each other and with the frame element **27**. By doing so, they mitigate the transfer of thermal energy from outside of the unit to the interiors **17** and **18** of the unit **10**.

[0121] Assembling the air handling unit **10** with the insulation panels **20** on the frame elements **25** and **27** described herein enables the frame elements **25** and **27** to be recessed well below the common outer surface **131** of the air handling unit **10**, thereby insulating the frame elements **25** and **27** from thermal transfer. Moreover, the gaskets **160** and sealants **162** prevent transmission of air between the environment and the interior **17** and **18**. Finally, the long assemblies of frame elements **25** enables construction of a continuous, single-compartment air handling unit **10**, improving its operational efficacy and efficiency.

[0122] A preferred embodiment is fully and clearly described above so as to enable one having skill in the art to understand, make, and use the same. Those skilled in the art will recognize that modifications may be made to the description above without departing from the spirit of the specification, and that some embodiments include only those elements and features described, or a subset thereof. To the extent that modifications do not depart from the spirit of the specification, they are intended to be included within the scope thereof.

Claims

1. An air handling unit comprising: a frame element; first and second insulation panels which confront the frame element at a juncture; the first insulation panel includes a first distal surface, a first proximal surface, a first intermediate surface extending between the first distal and proximal surfaces, and a first inner surface; the second insulation panel includes a second distal surface, a second proximal surface, a second intermediate surface extending between the second distal and proximal surfaces, and a second inner surface; and at the juncture: the second distal surface faces the first intermediate surface at an insulation seam; the second proximal surface faces in spaced

relation the first intermediate surface and the second intermediate surface faces in spaced relation the first proximal surface, thereby defining a closed void defined between the first intermediate surface, the first proximal surface, the second intermediate surface, and the second proximal surface; and the frame element occupies the closed void; wherein: a first air-impermeable seal is in the insulation seam; second air-impermeable seals are between the frame element and the first and second intermediate surfaces, respectively, and between the frame element and the first and second inner surfaces, respectively; and third air-impermeable seals are between the frame element and the first and second proximal surfaces, respectively.

2. The air handling unit of claim 1, wherein: the first distal surface is parallel to the first proximal surface, and the second distal surface is parallel to the second proximal surface; the first intermediate surface is normal to the first distal surface and to the first proximal surface, and the second intermediate surface is normal to the second distal surface and to the second proximal surface; and the first inner surface is normal to the first distal surface and to the first proximal surface, and the second inner surface is normal to the second distal surface and to the second proximal surface.

3. The air handling unit of claim 1, wherein: the first and second insulation panels are normal to each other; and at the juncture, the first insulation panel overlaps the second insulation panel.

4. The air handling unit of claim 1, wherein: the first air-impermeable seal in the insulation seam is a sealant extending along the insulation seam; the second air-impermeable seal between the frame element and the first intermediate surface comprises two gaskets each extending along a length of the frame element; the second air-impermeable seal between the frame element and the second intermediate surface comprises another two gaskets each extending along the length of the frame element; the second air-impermeable seal between the frame element and the first inner surface comprises another gasket extending along the length of the frame element; the second air-impermeable seal between the frame element and the second inner surface comprises yet another gasket extending along the length of the frame element; the third air-impermeable seal between the frame element and the first proximal surface is another sealant extending along a length of the frame element; and the third air-impermeable seal between the frame element and the second proximal surface is yet another sealant extending along the length of the frame element.

5. An air handling unit comprising: a frame element; first and second insulation panels which confront the frame element at a juncture; the first insulation panel includes a first distal surface, a first proximal surface, a first intermediate surface extending between the first distal and proximal surfaces, and a first inner surface; the second insulation panel includes a second distal surface, a second proximal surface, a second intermediate surface extending between the second distal and proximal surfaces, and a second inner surface; and at the juncture: the first distal surface faces the second distal surface at an insulation seam; the first intermediate surface is registered in alignment with the second intermediate surface; the first proximal surface faces the second proximal surface in spaced relation, thereby defining an open void between the first intermediate surface, the first proximal surface, the second intermediate surface, and the second proximal surface; and the frame element occupies the open void; wherein: a first air-impermeable seal is in the insulation seam; second air-impermeable seals are between the frame element and the first and second intermediate surfaces, respectively, and between the frame element and the first second inner surfaces, respectively; and third air-impermeable seals are between the frame element and the first and second proximal surfaces, respectively.

6. The air handling unit of claim 5, wherein: the first distal surface is parallel to the first proximal surface, and the second distal surface is parallel to the second proximal surface; the first intermediate surface is normal to the first distal surface and to the first proximal surface, and the second intermediate surface is normal to the second distal surface and to the second proximal surface; and the first inner surface is normal to the first distal surface and to the first proximal surface, and the second inner surface is normal to the second distal surface and to the second

proximal surface.

7. The air handling unit of claim 5, wherein: the first and second insulation panels are registered in alignment with each other; and at the juncture, the first insulation panel abuts the second insulation panel.

8. The air handling unit of claim 5, wherein: the first air-impermeable seal in the insulation seam is a sealant extending along the insulation seam; the second air-impermeable seal between the frame element and the first intermediate surface comprises two gaskets each extending along a length of the frame element; the second air-impermeable seal between the frame element and the second intermediate surface comprises another two gaskets each extending along the length of the frame element; the second air-impermeable seal between the frame element and the first inner surface comprises another gasket extending along a length of the frame element; the second air-impermeable seal between the frame element and the second inner surface comprises yet another gasket extending along the length of the frame element; the third air-impermeable seal between the frame element and the first proximal surface is another sealant extending a length of the frame element; and the third air-impermeable seal between the frame element and the second proximal surface is yet another sealant extending along the length of the frame element.

9. An air handling unit comprising: a housing constructed from a plurality of insulation panels embedded with frame elements; the frame elements include at least a first frame element and a second frame element joined together by an inline connector; each insulation panel includes an outer surface and an opposed inner surface, wherein the insulation panels all cooperate to form (1) a common outer surface of the air handling unit comprised of the outer surfaces of the insulation panels and (2) a common inner surface of the air handling unit comprised of the inner surfaces of the insulation panels; wherein the common inner surface bounds and defines an interior space of the air handling unit; and the frame elements are all inboard of the common outer surface, recessed and embedded within the insulation panels.

10. The air handling unit of claim 9, wherein the inline connector comprises: a middle section; a first mating portion extending from the middle section in a first direction to a first end; a second mating portion extending from the middle section in a second direction, opposite the first direction, to a second end; the first mating portion comprising a first major plug and a first minor plug, wherein the first major and minor plugs are configured to fit into sockets in the frame elements; the second mating portion comprising a second major plug and a second minor plug, wherein the second major and minor plugs are configured to fit into the sockets in the frame elements; and the middle section includes a generally rectangular prismatic body having a dimension larger than the first and second mating portions, and two flanges projecting outwardly away from the body.

11. The air handling unit of claim 10, wherein: the first major plug and the first minor plug are separated by a slender first gap; and the second major plug and the second minor plug are separated by a slender second gap.

12. The air handling unit of claim 10, wherein the flanges are proximate the first and second minor plugs.

13. The air handling unit of claim 9, wherein: the insulation panels include at least a first insulation panel and a second insulation panel confronting one of the frame elements at a juncture; the first insulation panel includes a first distal surface, a first proximal surface, a first intermediate surface extending between the first distal and proximal surfaces, and a first inner surface; the second insulation panel includes a second distal surface, a second proximal surface, a second intermediate surface extending between the second distal and proximal surfaces, and a second inner surface; and at the juncture: the second distal surface faces the first intermediate surface at an insulation seam; the second proximal surface faces in spaced relation the first intermediate surface and the second intermediate surface faces in spaced relation the first proximal surface, thereby defining a closed void between the first intermediate surface, the first proximal surface, the second intermediate surface, and the second proximal surface; and at least one of the first and second frame elements

occupies the closed void.

14. The air handling unit of claim 13, wherein: a first air-impermeable seal is in the insulation seam; second air-impermeable seals are between the frame element and the first and second intermediate surfaces, respectively, and between the frame element and the first and second inner surfaces, respectively; and third air-impermeable seals are between the frame element and the first and second proximal surfaces, respectively.

15. The air handling unit of claim 14, wherein: the first air-impermeable seal in the insulation seam is a sealant extending along the insulation seam; the second air-impermeable seal between the frame element and the first intermediate surface comprises two gaskets each extending along a length of the frame element; the second air-impermeable seal between the frame element and the second intermediate surface comprises another two gaskets each extending along the length of the frame element; the second air-impermeable seal between the frame element and the first inner surface comprises another gasket extending along the length of the frame element; the second air-impermeable seal between the frame element and the second inner surface comprises yet another gasket extending along the length of the frame element; the third air-impermeable seal between the frame element and the first proximal surface is another sealant extending along a length of the frame element; and the third air-impermeable seal between the frame element and the second proximal surface is yet another sealant extending along the length of the frame element.

16. The air handling unit of claim 13, wherein: the first distal surface is parallel to the first proximal surface, and the second distal surface is parallel to the second proximal surface; the first intermediate surface is normal to the first distal surface and to the first proximal surface, and the second intermediate surface is normal to the second distal surface and to the second proximal surface; and the first inner surface is normal to the first distal surface and to the first proximal surface, and the second inner surface is normal to the second distal surface and to the second proximal surface.

17. The air handling unit of claim 9, wherein: the insulation panels include at least a first insulation panel and a second insulation panel confronting one of the frame elements at a juncture; the first insulation panel includes a first distal surface, a first proximal surface, a first intermediate surface extending between the first distal and proximal surfaces, and a first inner surface; the second insulation panel includes a second distal surface, a second proximal surface, a second intermediate surface extending between the second distal and proximal surfaces, and a second inner surface; and at the juncture: the first distal surface faces the second distal surface at an insulation seam; the first intermediate surface is registered in alignment with the second intermediate surface; the first proximal surface faces the second proximal surface in spaced relation, thereby defining an open void between the first intermediate surface, the first proximal surface, the second intermediate surface, and the second proximal surface; and at least one of the first and second frame elements occupies the open void.

18. The air handling unit of claim 17, wherein: a first air-impermeable seal is in the insulation seam; second air-impermeable seals are between the frame element and the first and second intermediate surfaces, respectively, and between the frame element and the first and second inner surfaces, respectively; and third air-impermeable seals are between the frame element and the first and second proximal surfaces, respectively.

19. The air handling unit of claim 18, wherein: the first air-impermeable seal in the insulation seam is a sealant extending along the insulation seam; the second air-impermeable seal between the frame element and the first intermediate surface comprises two gaskets each extending along a length of the frame element; the second air-impermeable seal between the frame element and the second intermediate surface comprises another two gaskets each extending along the length of the frame element; the second air-impermeable seal between the frame element and the first inner surface comprises another gasket extending along a length of the frame element; the second air-impermeable seal between the frame element and the second inner surface comprises yet another

gasket extending along the length of the frame element; the third air-impermeable seal between the frame element and the first proximal surface is another sealant extending a length of the frame element; and the third air-impermeable seal between the frame element and the second proximal surface is yet another sealant extending along the length of the frame element.

20. The air handling unit of claim 17, wherein: the first distal surface is parallel to the first proximal surface, and the second distal surface is parallel to the second proximal surface; the first intermediate surface is normal to the first distal surface and to the first proximal surface, and the second intermediate surface is normal to the second distal surface and to the second proximal surface; and the first inner surface is normal to the first distal surface and to the first proximal surface, and the second inner surface is normal to the second distal surface and to the second proximal surface.
